

Slide 4

Achievements

The proper use of git

The screenshot displays a Jira project board for 'ADS MT' in a dark theme. The interface includes a top navigation bar with the user 'zin-lin' and a search bar. Below the navigation bar, the project name 'ADS MT' is shown, followed by tabs for different views: Backlog, Priority board, Team items, Roadmap, In review, My items, and a '+ New view' button. A filter bar allows searching by keyword or field, with 'Discard' and 'Save' buttons. The main area is a Kanban board with five columns: 'Backlog' (3 items), 'Ready' (1 item), 'In progress' (1 item), 'In review' (0 items), and 'Done' (0 items). Each column has a description and an 'Add item' button at the bottom. The 'Backlog' column contains three items: 'vehicle_control_package #2' (P1, S), 'path_planner_package #2' (P2), and 'lidar_package #1' (P2, L). The 'Ready' column contains one item: 'vehicle_control_package #1' (P1). The 'In progress' column contains one item: 'path_planner_package #1' (P1). The 'In review' and 'Done' columns are currently empty.

zlin-lin / Projects / ADS MT

ADS MT

Backlog Priority board Team items Roadmap In review My items + New view

Filter by keyword or by field

Discard Save

Backlog 3 / 5 Estimate: 0

This item hasn't been started

- vehicle_control_package #2
Create Connection with control software
P1 S
- path_planner_package #2
Create integration with annex msgs
P2
- lidar_package #1
initiate lidar software
P2 L

+ Add item

Ready 1 Estimate: 0

This is ready to be picked up

- vehicle_control_package #1
vehicle control values
P1

+ Add item

In progress 1 / 3 Estimate: 0

This is actively being worked on

- path_planner_package #1
Create Vehicle Control For a PurePursuit Controller
P1

+ Add item

In review 0 / 5 Estimate: 0

This item is in review

+ Add item

Done 0 Estimate: 0

This has been completed

+ Add item

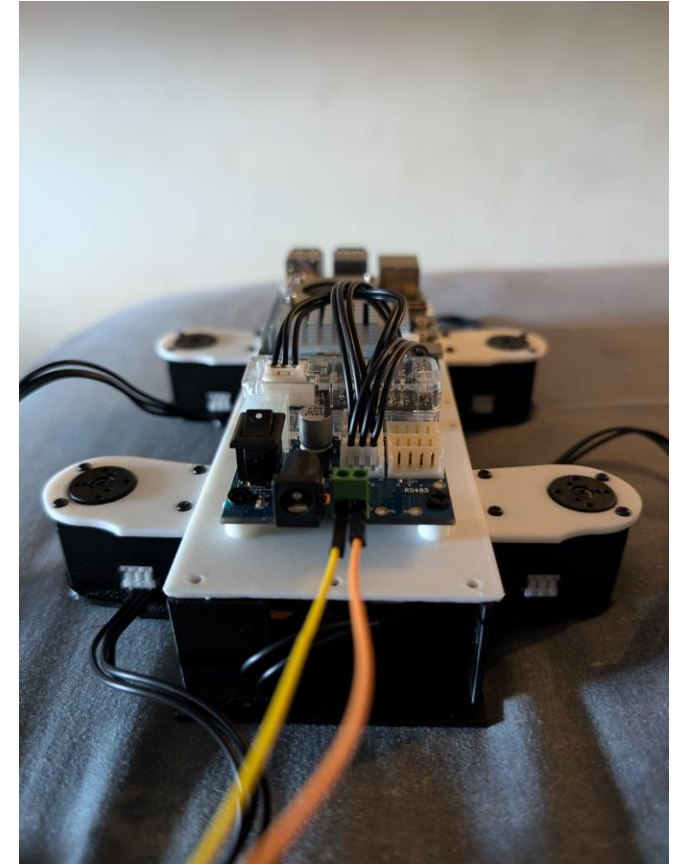
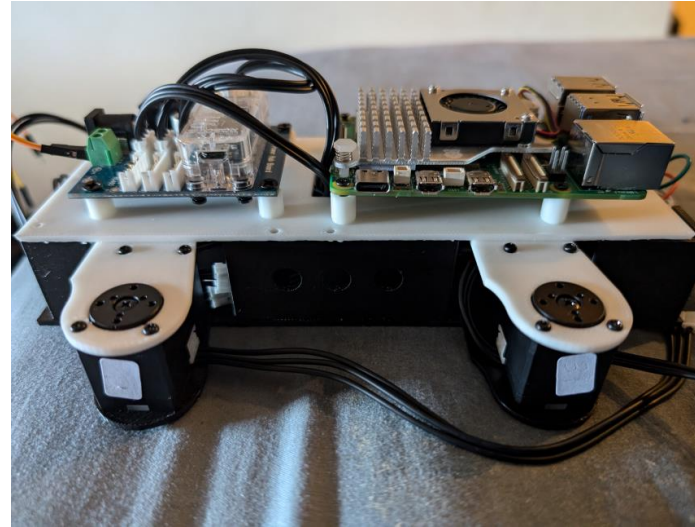
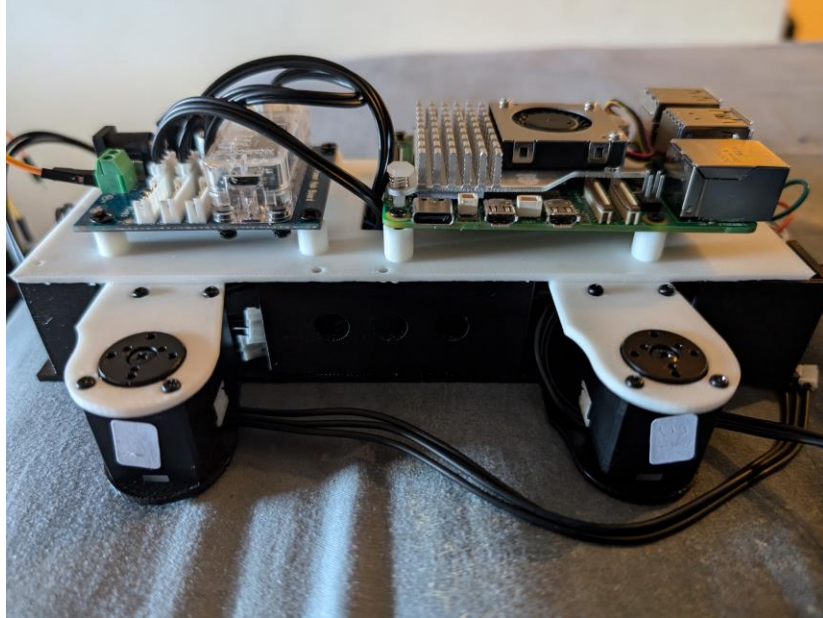
The proper use of issue branch and labelling

The screenshot displays a GitHub issue page for the repository 'zin-lin / Projects / ADS MT'. The issue title is 'Create Connection with control software #2', which is marked as 'Open' and is linked to the branch 'zin-lin/vehicle_control_package'. The issue is public and has a 'vehicle control' label.

The left sidebar shows the project's backlog with three items: 'vehicle_control_package #2' (Priority 1, Size S), 'path_planner_package #2' (Priority 2), and 'lidar_package #1' (Priority 2, Size L). The main content area shows the issue's history, including its creation by 'zin-lin' last week, its assignment to 'zin-lin', its conversion from a draft, and the addition of the 'vehicle control' label. A comment section at the bottom allows for adding new comments.

The right sidebar provides additional context, including the assignee 'zin-lin', the project 'ADS MT' (Backlog), and various metadata such as priority (P1), size (S), and dates. It also includes a 'Development' section with a link to create a branch for the issue and a 'Notifications' section with an 'Unsubscribe' button.

Construction Process



The main chassis is complete with battery insertion and wirework and is tested.

- The assembly is waiting upon the arrival of horn spacers being printed
- A new wheel design has been made
- Assembly should be complete next week

Future Milestones

- Get Literature Review done before the 15th of December (December 6th - 15th)
- Get the robot built fully (November 15th- 20th)
- Get the URDF done and tested (December 15th - 20th)
- Get the control software done (December 15th – 30th)
- (Maybe) SLAM research (January)
- By January start writing part of the methodology
- Next 3 months will be data collection, logging etc.
- Get more modules installed on the robot
- (Maybe) train the robot to walk on a simulator.

More papers on the 3rd theme

- Zhou, Z., Zhang, C., Li, C., Zhang, Y., Shi, Y., & Zhang, W. (2024). A tightly-coupled LIDAR-IMU SLAM method for quadruped robots. *Measurement and Control (London)*, 57(7), 1004–1013.
<https://doi.org/10.1177/00202940231224593>
- Cheng, J., Zhang, M., Ma, L., Chen, H., Gan, Y., & Huang, D. (2024). A Hybrid-Dimensional Laser SLAM Framework for Indoor Quadruped Inspection Robots. *IEEE Sensors Journal*, 24(10), 16935–16942.
<https://doi.org/10.1109/JSEN.2024.3382336>
- He, J., Shao, J., Sun, G., Cheong, S. A., & Siew Ann Cheong. (2023). The Quadruped Robot Uses the Trajectory Planned by DIACO to Complete the Obstacle Avoidance Task. *Complexity (New York, N.Y.)*, 2023, 1–14. <https://doi.org/10.1155/2023/8128847>
- Halder, S., Afsari, K., Chiou, E., Patrick, R., & Hamed, K. A. (2023). Construction inspection & monitoring with quadruped robots in future human-robot teaming: A preliminary study. *Journal of Building Engineering*, 65, 105814-. <https://doi.org/10.1016/j.jobbe.2022.105814>
- Amerttet, S., Gebresenbet, G., & Alwan, H. M. (2024). Optimizing the performance of a wheeled mobile robots for use in agriculture using a linear-quadratic regulator. *Robotics and Autonomous Systems*, 174, 104642-. <https://doi.org/10.1016/j.robot.2024.104642>
- Zijlstra, G., Aplin, K. L., & Hunt, E. R. (2024). Exploring the Use of Terrestrial Robots for Atmospheric Electricity Measurement. *Journal of Physics. Conference Series*, 2702(1), 12006-.
<https://doi.org/10.1088/1742-6596/2702/1/012006>